
MH-SIGNALS: Multi-Signal Mental Health Detection and Grounded Response Generation

Abstract

Online mental health communities contain large volumes of user-generated posts expressing distress, advice-seeking behavior, and crisis-related communication. Manual moderation and response at scale is difficult due to both volume and sensitivity. We propose MH-SIGNALS, a unified framework for multi-signal mental health detection and retrieval-augmented response generation. Given a forum post, our system predicts communicative intent and concern level (Low, Medium, High). For elevated-risk posts, it generates empathetic responses grounded in a curated knowledge base of counselor-style replies. We benchmark MiniLM with Logistic Regression, DistilRoBERTa with LoRA, and RoBERTa with LoRA under a reproducible pipeline and evaluate responses using semantic similarity, retrieval grounding, and manual safety auditing.

1 Introduction

The growing presence of mental health discussions in online communities has created both an opportunity and a challenge for researchers and practitioners. Platforms such as Reddit’s mental health subreddits host thousands of posts daily from individuals experiencing varying degrees of psychological distress; however, the volume and sensitivity of this content make consistent human moderation and response nearly impossible at scale. Automated tools that can accurately identify the nature and severity of these posts and respond to them meaningfully have significant potential to support both moderators and the individuals seeking help.

In this project, we address this problem as a combined signal detection and response generation task. Given a raw text post from an online mental health forum, our system predicts two signals: the user’s intent (e.g., expressing distress, seeking advice, sharing an experience) and the concern level of the post (Low, Medium, or High). For elevated concern posts, the system additionally generates an empathetic, counselor-style reply grounded in a curated knowledge base of real support responses. The desired output is therefore structured, containing intent tags, a concern level, a generated reply with citations, and crisis resources where appropriate.

We chose this problem because existing moderation approaches are largely keyword-based and fail to capture the nuanced language patterns present in mental health discourse. General-purpose language models, while fluent, lack the domain grounding and safety constraints necessary for this context. MH-SIGNALS addresses both gaps by benchmarking multiple classification strategies within a reproducible pipeline and pairing them with a retrieval-augmented generation layer designed specifically for sensitive, high-stakes interactions.

2 Related Work

2.1 Mental Health Detection from Social Media

Prior research has demonstrated that social media language can reveal indicators of mental health conditions. De Choudhury et al. [2] showed that linguistic patterns on Twitter could predict depression

before clinical diagnosis. Similarly, Coppersmith et al. [1] applied supervised learning to detect mental health signals from user posts.

Transformer-based models such as BERT [3] and RoBERTa [9] significantly improved contextual modeling. These models have been applied to suicide risk detection and distress classification [10]. However, most prior work focuses on binary classification. Our work extends this by modeling structured multi-signal outputs including intent and severity level.

2.2 Parameter-Efficient Fine-Tuning

Fine-tuning large transformers is computationally expensive. LoRA [5] introduces low-rank adaptation matrices into attention layers, enabling efficient model adaptation. Adapter-based approaches [4] similarly reduce parameter cost. We evaluate LoRA-based fine-tuning to balance efficiency and performance in mental health classification.

2.3 Retrieval-Augmented Generation and Safety

Retrieval-Augmented Generation (RAG) [8] combines dense retrieval with generative models to improve grounding and reduce hallucination. Dense retrieval methods such as DPR [7] enable efficient semantic search.

Grounding is critical in high-risk domains such as mental health. Large language models are known to hallucinate [6], making safety-aware evaluation essential. Our system integrates retrieval grounding with structured safety checks and manual auditing.

3 Dataset

For this project, we use a dataset of online mental health support group posts collected from publicly available discussion platforms. The dataset consists of user-written posts where individuals describe their emotions, personal struggles, anxiety, depression, academic stress, relationship issues, and other mental health related concerns.

The raw data is stored in CSV format and contains the post text along with minimal metadata. Since the original dataset did not include structured labels aligned with our task, we created additional annotations to support our classification objectives.

We enriched the dataset with two types of labels:

Intent – This captures the primary purpose of the post, such as emotional expression, seeking advice, venting, or crisis related communication.

Concern Level – This represents the severity of the situation described in the post and is categorized as Low, Medium, or High.

The labeling was performed using a rule-based tagging framework designed specifically for mental health language patterns. The tagging rules rely on commonly observed linguistic cues, distress indicators, and crisis related phrases. After tagging, we reviewed label distributions to ensure reasonable class balance.

The final processed dataset is stored as a cleaned and tagged CSV file. We created fixed train, validation, and test splits to ensure consistent evaluation across all experiments.

In addition to the classification dataset, we constructed a small knowledge base consisting of counselor-style responses. These responses are used in the retrieval component of our system to generate grounded replies. The knowledge base is indexed using dense embeddings to allow efficient similarity-based retrieval.

4 Approach

4.1 Input Processing

User posts are preprocessed and cleaned to standardize formatting while preserving emotional content. The labeled dataset supports prediction of Intent and Concern Level.

4.2 Signal Detection

We benchmark three approaches:

MiniLM + Logistic Regression: Sentence embeddings generated using MiniLM are classified using Logistic Regression as a lightweight baseline.

DistilRoBERTa + LoRA: DistilRoBERTa is fine-tuned using LoRA for parameter-efficient adaptation.

RoBERTa-base + LoRA: A stronger transformer backbone is fine-tuned using LoRA to capture deeper contextual cues.

All models are trained and evaluated using identical splits.

4.3 Retrieval Pipeline

We build a retrieval-augmented generation pipeline that retrieves relevant counselor-style snippets from a curated Knowledge Base using FAISS-based embedding search. Retrieved snippets serve as grounding evidence.

4.4 Grounded Generation

A Flan-T5 generator combines the original post with retrieved snippets to generate a supportive, empathetic response grounded in evidence rather than generic text. Crisis resources are appended for High concern cases.

5 Evaluation Plan

We evaluate the system across three dimensions: classification performance, response generation quality, and safety compliance.

5.1 Classification Performance

We compute:

- Macro-F1 Score (primary metric)
- Per-class Recall and Precision
- Confusion Matrix
- High-Risk Recall

Macro-F1 ensures equal weight across classes. High-risk recall directly measures detection of crisis-level content. A confusion matrix heatmap is generated to diagnose systematic misclassifications.

5.2 Response Generation Quality

We compute:

- BERTScore (semantic similarity)
- Retrieval Recall@3

BERTScore evaluates contextual similarity between generated and reference responses. Retrieval Recall@3 measures grounding effectiveness.

5.3 Safety and Manual Audit

We manually evaluate 50 test examples using binary Pass/Fail criteria:

- Crisis Resource Trigger

- Clinical Safety (no medical advice)
- Tone Appropriateness

This ensures ethical compliance beyond automatic metrics.

References

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